Adam von Kannewurff

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Education

Georgia Institute of Technology — Remote
Master of Science in Computer Science — *January 2023 – December 2025*
Coursework: *Distributed Computing, Operating Systems, Databases, High-Performance Computing*

Virginia Polytechnic Institute and State University — Blacksburg, VA
Bachelor of Science in Industrial and Systems Engineering — *September 2014 – May 2019*

Work Experience

Senior Software Engineer — June 2019 – Present
KPMG — *New York, NY*

- Orchestrated the consolidation of multiple legacy personnel systems into a unified Salesforce platform; led 5 cross-functional teams to deliver 6 applications, scaling capacity to 300,000 total users and 5,000+ DAU
- Architected a fleet optimization engine in an IL4 environment using time-series and simulation ML models; reduced planning cycles by 75% and provided stakeholders with predictive insights that replaced reactive legacy workflows
- Engineered the infrastructure for a multi-million dollar IoT smart warehouse, implementing hyper-converged virtualization to process 1,000+ RFID scans per second via 100+ physical devices using high-availability networking
- Led 3 teams through enterprise ServiceNow implementation, delivering 5 Healthcare HR applications to 1,000 DAU
- Developed full-stack MVP accelerator, reducing "Time-to-MVP" time by 50%, contributing to multiple contracts wins
- Supported migration of CDKTF in-house IaC library to native HCL, enabling continued operations of infrastructure services

Federal Advisory Intern — June 2018 – August 2018
KPMG — *McLean, VA*

- Developed framework to facilitate pursuance of software contracts, reducing proposal preparation time by 40%

Achievements

HackOMSCS Winner | *"AI Awakening" Category* — May 2024

- Engineered a "smart pantry" solution in 24 hours that utilizes computer vision to identify ingredients from user-submitted images and generates personalized recipes via machine learning integration
- Architected a full-stack cloud-native application by integrating Azure AI Vision for image analysis and open-source ML model for recipe generation, delivering a responsive end-to-end user experience under strict time constraints

Projects

Sharded Key-Value Store | *Java, DSLabs*

- Engineered a linearly consistent, sharded K/V store using a Multi-Paxos consensus engine to ensure high availability and state machine replication across distributed nodes
- Implemented a Shard Controller to manage dynamic data rebalancing and configuration changes, utilizing a pull-based state transfer mechanism to handle node joins and failures without data loss

Paxos | *Java, DSLabs*

- Developed a complete Multi-Paxos implementation from scratch, optimizing for performance via log compaction and a heartbeat-based leader election to minimize latency
- Designed robust failure recovery logic to handle network partitions, message drops, and dueling proposers, verified through rigorous adversarial model checking and state-space exploration

Certifications

Salesforce: Architect: Data, Development Lifecycle and Deployment, Sharing and Visibility, Application, +8 other certifications
Other: CompTIA Security+, Scrum Master, ServiceNow Administrator

Technical Skills

Languages: Java, C/C++, JavaScript/TypeScript, Python, SQL (Postgres), HTML/CSS
Cloud & DevOps: AWS, Azure, Kubernetes, Docker, Terraform (HCL/CDKTF), GitLab CI/CD, Splunk
Frameworks: Node.js, FastAPI, React, Vue, Vitest, Jest